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Where Tech with Intelligence Meets Soil

1 Cotton Weed Identification and Management

1.1 Agronomic Significance, Crop-Weed Competition, and Yield Losses

- **Agronomic vulnerability of cotton to weed pressure:** Cotton (*Gossypium* spp.) is inherently sensitive to weed competition during early vegetative growth. The crop exhibits slow initial growth and wide row-to-row spacing (67.5 to 108 cm), leaving large exposed soil inter-row areas. Under open sunlight, high temperatures, and moist seedbed conditions, native weed seeds germinate rapidly, overtopping young cotton seedlings within 10 to 14 days.
- **Critical Period of Crop-Weed Competition (CPC):** The first 8 to 9 weeks after sowing (0 to 60 Days After Sowing) represent the critical period of crop-weed competition. Weed interference occurring during this 60-day window causes irreversible physiological stunting and yield reductions ranging from 40% to 85%. Weed emergence after canopy closure (beyond 60 to 75 DAS) causes minimal direct yield loss, but late-season weeds interfere with manual boll picking and contaminate lint.
- **Resource competition dynamics:** Fast-growing annual grasses and broadleaf weeds compete aggressively for soil moisture, applied macro- and micronutrients, and solar radiation. Weeds like *Trianthema portulacastrum* accumulate nitrogen and potassium at twice the rate of cotton seedlings, starving the crop's shallow lateral roots during early squaring.
- **Indirect damage as alternate pest and disease reservoirs:** Uncontrolled weeds serve as primary overwintering hosts and nutritional bridges for destructive insect pests and plant pathogens. Malvaceous weeds such as *Abutilon indicum* and *Sida cordifolia* harbor whiteflies (*Bemisia tabaci*) and act as primary reservoirs for Cotton Leaf Curl Virus (CLCuD). Weeds such as *Parthenium hysterophorus* support tobacco streak virus (TSV), thrips, and invasive mealybugs (*Phenacoccus solenopsis*).

1.2 Classification and Taxonomic Identification of Major Weed Flora

- **Major Sedge Species:**
  - *Cyperus rotundus* (Motha / Purple nutsedge): Perennial sedge with dark-green triangular stems, three-ranked basal leaves, and branched underground rhizome systems terminating in starch-rich basal tubers. Highly persistent;

shoots regenerate repeatedly from subterranean tubers within 4 to 6 days after manual hoeing.

- *Cyperus iria* (Yellow nutsedge / Rice flat sedge): Annual sedge with fibrous roots and yellow-brown spikelets, thriving in heavy or saturated low-lying cotton fields.

- **Major Monocot Grass Weeds:**
  - *Cynodon dactylon* (Doob ghas / Khabbal ghas / Bermudagrass): Hard, creeping perennial grass spreading via tough surface stolons and deep underground rhizomes. Forms dense vegetative mats along crop rows, depleting surface moisture.
  - *Dactyloctenium aegyptium* (Gurta ghas / Crowfoot grass): Tufted annual grass with characteristic digitate spikes resembling a bird's foot. Germinates in flushes following early irrigations and rainstorms.
  - *Echinochloa crus-galli* and *Echinochloa colona* (Swank / Deela / Barnyard grass): Rapidly growing, tall annual grasses with flat leaf blades and purplish awned or awnless panicles. Vigorously competes with cotton for nitrogen and light during vegetative elongation.
  - *Digitaria sanguinalis* and *Digitaria marginata* (Crabgrass): Prostrate or decumbent annual grasses rooting at lower stem nodes, thriving under irrigated conditions.
  - *Eleusine indica* (Goosegrass): Tough, erect annual grass with strong fibrous roots that make mature plants difficult to uproot manually.
  - *Dinebra retroflexa* (Viper grass) and *Panicum repens* (Torpedo grass): Invasive grass species infesting black clay Vertisols and canal-irrigated basins.

- **Major Dicot Broadleaf Weeds:**
  - *Trianthema portulacastrum* (Itsit / Giant pigweed / Carpet weed): Prostrate, succulent, branched annual broadleaf weed with pinkish-white solitary flowers and opposite obovate fleshy leaves. The dominant early-season weed across the North Zone, forming dense green carpets within 7 to 10 days of sowing that smother emerging cotton seedlings.
  - *Digera arvensis* / *Digera muricata* (Tandla / False amaranth): Erect annual broadleaf herb with alternate ovate leaves and pinkish-white axillary spikes. Competes aggressively in sandy loam soils.
  - *Amaranthus viridis* (Chaulai / Pigweed): Vigorous erect annual broadleaf weed with deep taproots and dense terminal green spikes, producing thousands of long-lived seeds per plant.
  - *Cucumis trigonus* (Chibber bel) and *Ipomoea* spp. (Lapeta bel / Morning glory): Creeping and twining annual vines that climb up cotton stalks, binding main stems and sympodial branches together, deforming plant architecture and interfering with mechanical operations and picking.
  - *Cleome viscosa* (Hulhul / Sticky spider-flower): Sticky, glandular, erect annual herb with yellow flowers and elongated siliques.

  - *Tribulus terrestris* (Bhakhkhra / Puncture vine): Prostrate trailing herb with sharp, woody, multi-spined burrs that injure farm workers during manual weeding.
  - *Euphorbia microphylla* and *Euphorbia hirta* (Dhudhi / Spurge): Prostrate herbs with milky latex that form low mats across inter-row spaces.
  - *Xanthium strumarium* (Burdock datura / Cocklebur): Coarse annual herb with large spiny burs, serving as a primary host for whiteflies and CLCuD.
  - *Abutilon indicum* (Indian mallow / Kanghi buti) and *Parthenium hysterophorus* (Congress grass): Tall perennial and annual weeds that establish along field borders, irrigation ditches, and wasteland, acting as year-round pest and viral reserves.

1.3 Preventive Weed Management and Cultural Sanitation Measures

- **Deep Summer Mouldboard Ploughing and Solar Desiccation:**
  - Execute deep summer ploughing during the hottest months (May to June) immediately following the harvest of preceding Rabi crops using a tractor-drawn Mouldboard (MB) plough.
  - Deep tillage inverts the topsoil layer, bringing underground rhizomes, tubers, and rootstocks of perennial weeds (*Cyperus rotundus*, *Cynodon dactylon*) to the surface.
  - Leave inverted clods exposed to direct sunlight for at least 2 to 3 weeks. High surface temperatures (>45°C) bake and desiccate vegetative propagules, depleting the subterranean weed bank by over 70%.
- **Eradication of Alternate Host Weeds on Field Perimeters:**
  - Thoroughly clear and destroy volunteer weeds along field bunds, water channels, road verges, and adjoining fallow land prior to sowing.
  - Uproot and burn or deeply bury species of *Abutilon*, *Sida*, *Parthenium*, *Xanthium*, *Achyranthes*, and *Datura* before they produce seeds.
  - Eliminating these boundary weeds destroys the primary early-season feeding zones of whitefly vectors, cutting off the transmission chain for Cotton Leaf Curl Virus (CLCuD) into newly emerging cotton plots.
- **Use of Certified, Clean Planting Seed:** Ensure planting seed is certified, acid-delinted, and free from weed seed contaminants, preventing the introduction of foreign or noxious weed species into clean fields.
- **Well-Decomposed Organic Manure Usage:** Farmyard Manure (FYM) or compost applied during land preparation must be thoroughly decomposed in hot compost pits (> 55°C). Applying fresh or partially decomposed manure introduces thousands of viable weed seeds directly into the seedbed.

1.4 Mechanical Weed Control and Tractor-Operated Interculti-vation

- **Timing of Intercultural Operations:**
  - The cotton crop must be maintained completely weed-free for the first 8 to 9 weeks after sowing through timely mechanical hoeing and intercultivation.
  - The first intercultural operation or manual hoeing should be conducted 18 to 20 days after seedling emergence (before applying the first post-sowing irrigation) to uproot early flushes of germinating weeds.
  - Repeat mechanical intercultivation 2 to 3 times at 15 to 20-day intervals depending on weed emergence and soil crust formation.
- **Animal- and Tractor-Drawn Inter-Row Machinery:**
  - *Tractor-Operated Inter-Row Cultivator / Triphali:* Spring-loaded tine tillers or three-tined cultivators (Triphali) operated between 67.5 to 100 cm spaced rows. Effectively uproots inter-row weeds, pulverizes the surface crust, and establishes a loose dust mulch that conserves subsoil moisture.
  - *Tractor-Operated Inter-Row Rotary Weeder:* Power-take-off (PTO) driven rotary blades that chop and incorporate weed biomass into the inter-row soil profile during early vegetative stages (20 to 45 DAS).
  - *Engine-Operated Walking Power Weeders:* Compact motorized rotary tillers suitable for smallholder farms and medium-width rows.
  - *Animal-Operated Triphali / Blade Harrows:* Bullock- or camel-drawn three-tine cultivators or blade harrows (Bakhar) commonly utilized across North, Central, and South Indian cotton tracts.
- **Intra-Row Manual Weeding (Kasola):** Mechanical tractor cultivators cannot operate close to plant crowns without risking seedling damage. Hand weeding and hoeing along intra-row spaces must be performed using long-handled iron hoes (Kasola / Wheel hand hoe) to remove stubborn weeds growing between plants within rows.
- **Operational Restriction after Fruiting:** Mechanical intercultivation using tractor-drawn cultivators or animal-drawn implements must be discontinued once plants initiate sympodial branching and squaring (beyond 60 to 70 DAS). Late tillage damages shallow lateral feeder roots, dislodges low-hanging fruiting branches, and causes mechanical flower and square drop.

1.5 Ecological Suppression: Cover Cropping, Intercropping, and Mulching

- **Short-Duration Leguminous Green Manure Intercropping:**
  - Sowing short-duration legumes between wide cotton rows provides ground canopy cover, outcompeting emerging weed seedlings for light and space.

  - Recommended intercrops include Cowpea (*Vigna unguiculata*), Green gram / Moong (*Vigna radiata*), Black gram (*Vigna mungo*), and Soybean (*Glycine max*) planted in a 1:1 or 1:2 row ratio.
  - *Biomass Mulching Soil Health:* Legumes are incorporated into the soil using a cultivator after 30 to 45 days of growth (before flowering). In-situ incorporation provides green manure mulch, adds organic matter, suppresses late weed flushes, and fixes atmospheric nitrogen.
  - *Whitefly Caution in North Zone:* Intercropping green gram (Moong) in cotton is prohibited in whitefly-endemic tracts of Punjab, Haryana, and Rajasthan, as Moong acts as a breeding nursery for whitefly vectors and CLCuD. In these areas, use non-host cover crops or organic mulches instead.

- **Polyethylene and Organic Mulching Systems:**
  - *Black Polyethylene Mulch:* Laying 30-micron black or silver-on-black polyethylene film along raised planting beds before sowing blocks photosynthetically active radiation (PAR), suppressing weed seed germination while conserving soil moisture.
  - *Organic Crop Straw Mulching:* Spreading a 5 to 7 cm thick layer of dry wheat straw, paddy straw, or shredded cotton stalks across inter-row spaces suppresses weed emergence, buffers root-zone temperatures against summer heat, and decomposes into organic carbon.

1.6 Chemical Weed Management: Pre-Plant Incorporation and Pre-Emergence Herbicides

- **Pre-Plant Soil Incorporation (PPI) Protocols:**
  - *Fluchloralin 45% EC:* Apply @ 2.0 to 2.68 L/ha (0.9 to 1.2 kg a.i./ha) dissolved in 500 to 800 L of water.
  - *Application Protocol:* Spray uniformly onto the prepared seedbed prior to sowing and incorporate into the top 3 to 5 cm soil layer using a light disc harrow or cultivator within 2 to 3 hours of spraying.
  - *Target Spectrum:* Inhibits cell division in germinating weed seeds; controls annual grass weeds (*Echinochloa*, *Digitaria*, *Eleusine*, *Cenchrus*) and small-seeded broadleaf weeds (*Amaranthus*, *Portulaca*, *Trianthema*, *Cleome*, *Chenopodium*).
  - *Trifluralin 48% EC:* Apply @ 2.0 to 2.5 L/ha (0.96 to 1.20 kg a.i./ha) as a pre-sowing soil incorporation in moist soil.

- **Pre-Emergence (PE) Soil-Active Herbicides:**
  - *Pendimethalin 30% EC:* Apply @ 2.5 to 4.165 L/ha (0.75 to 1.25 kg a.i./ha) dissolved in 500 to 700 L of water per hectare. Spray uniformly across the soil surface within 24 to 48 hours of sowing (before crop and weed emergence).
  - *Pendimethalin 38.7% CS (Capsule Suspension):* Advanced micro-encapsulated formulation applied @ 1.5 to 1.75 L/ha (677 g a.i./ha) in 500 L of water. The micro-capsules release the active ingredient slowly, extending soil residual weed

control up to 40 to 50 days while reducing photodecomposition losses under intense summer heat.

- *Alachlor 50% EC / 10% GR:* Apply Alachlor 50% EC @ 4.0 to 5.0 L/ha (2.0 to 2.5 kg a.i./ha) in 250 to 500 L of water, or broadcast Alachlor 10% granules @ 20 to 25 kg/ha on moist soil within 2 days of sowing to control annual grasses (*Dactyloctenium*, *Echinochloa*) and dicot weeds.
- *Diuron 80% WP:* Apply @ 1.0 to 2.2 kg/ha (0.75 to 1.5 kg a.i./ha) dissolved in 625 L of water as a pre-emergence blanket spray on medium to heavy black clay soils.

- **Field Conditions for Pre-Emergence Herbicide Efficacy:**
  - The seedbed must be level, firm, clod-free, and pulverized, free from coarse unchopped crop stubble.
  - Adequate moisture in the top 5 cm soil profile is mandatory at application time; herbicides sprayed onto dry, dusty soil fail to activate and volatilize under solar radiation.

1.7 Chemical Weed Management: Early Post-Emergence and Directed Sprays

- **Early Post-Emergence Selective Herbicides (Over-the-Top / Semi-Directed):**
  - *Pyrithiobac-Sodium 10% EC:* Selective post-emergence broadleaf herbicide applied @ 625 to 750 mL/ha (62.5 to 75 g a.i./ha) dissolved in 500 L of water. Apply at 20 to 30 DAS when broadleaf weeds are at the 2 to 4-leaf stage. Controls *Trianthema portulacastrum*, *Amaranthus viridis*, *Digera arvensis*, *Chenopodium album*, and *Celosia argentea* without injuring cotton seedlings.
  - *Quizalofop-Ethyl 5% EC:* Selective post-emergence graminicide applied @ 1000 mL/ha (50.5 g a.i./ha) dissolved in 500 L of water. Apply when annual grasses are at the 2 to 4-leaf stage (20 to 30 DAS). Highly effective against *Echinochloa crus-galli*, *E. colona*, *Dactyloctenium aegyptium*, *Digitaria marginata*, and *Dinebra retroflexa*.
  - *Fenoxaprop-p-Ethyl 9.3% w/w EC (9% w/v):* Selective grass weed herbicide applied @ 750 mL/ha (67.5 g a.i./ha) in 375 to 500 L of water at 20 to 25 DAS. Absorbed rapidly through weed foliage, halting grass growth within 48 hours.
  - *Ready-Mix Combination (Pyrithiobac-Sodium 6% + Quizalofop-Ethyl 4% 10 MEC):* Broad-spectrum post-emergence herbicide applied @ 1250 mL/ha dissolved in 375 L of water after the first post-sowing irrigation (20 to 30 DAS). Controls both annual broadleaf weeds and grassy monocots in a single application.

- **Directed Post-Emergence Non-Selective Herbicides (Inter-Row Hooded Spraying):**

  - *Target Stage:* Executed at 45 to 60 DAS to control mixed weed growth between rows where mechanical tillage cannot operate.
  - *Paraquat Dichloride 24% SL:* Contact herbicide applied @ 1.25 to 2.0 L/ha (0.3 to 0.5 kg a.i./ha) in 500 L of water. Provides quick burn-down of emergent sedges, grasses, and broadleaf weeds.
  - *Glufosinate Ammonium 13.5% SL (15% w/v):* Contact, broad-spectrum post-emergence herbicide applied @ 2.5 to 3.0 L/ha (375 to 450 g a.i./ha) in 500 L of water. Effectively burns down *Cyperus rotundus*, *Cynodon dactylon*, *Echinochloa*, and broadleaf species.
  - *Protective Spray Hood Requirement:* Non-selective herbicides (Paraquat, Glufosinate) must be applied strictly as directed inter-row sprays using sprayers fitted with protective hoods. Herbicide droplets or spray drift coming into contact with green cotton stems, leaves, or fruiting branches causes severe scorching and necrosis.

1.8 Herbicide Application Precision, Equipment Calibration, and Safety Stewardship

- **Sprayer Machinery and Nozzle Selection:**
  - Herbicides must be applied using knapsack sprayers, battery-operated sprayers, or tractor-mounted boom sprayers equipped with flat-fan or flood-jet (deflector) nozzles.
  - Hollow-cone nozzles (designed for insecticidal foliar misting) must never be used for herbicide applications because they create fine aerosol droplets that drift onto non-target crops and fail to deliver uniform soil coverage.
- **Spray Volume and Water Quality Standards:**
  - Maintain recommended water volumes: 500 to 700 L/ha for pre-emergence soil sprays, and 375 to 500 L/ha for post-emergence foliar treatments.
  - Use clean, clear water free from suspended mud, organic silt, or clay particles. Herbicides such as Paraquat and Glyphosate bind tightly to suspended soil colloids, causing chemical deactivation.
- **Sprayer Calibration and Delivery Height:**
  - Calibrate nozzle flow rate, operating pressure (1.5 to 2.0 bar), and operator walking speed (1.0 m/s) prior to field application to ensure exact dosage delivery.
  - Maintain spray boom height at 45 to 50 cm above the soil surface or weed canopy, ensuring a 30% spray pattern overlap between adjacent nozzles for uniform distribution.
- **Environmental Constraints during Spray Operations:**
  - Spray during calm morning or late afternoon hours when wind speeds are below 5 to 8 km/h. Avoid spraying during high winds (>10 km/h) to prevent drift onto adjoining sensitive crops (vegetables, pulses, orchards).

  - Never apply pre-emergence herbicides when heavy rainfall is imminent; excessive rainwater leaches herbicides into the seed placement zone (3 to 5 cm depth), causing crop seedling phytotoxicity.

- **Herbicide Resistance Management (HRM) Stewardship:**
  - Avoid repeated, continuous use of herbicides sharing the same chemical mode of action across multiple seasons.
  - Rotate ALS-inhibiting herbicides (Pyrithiobac-sodium) with ACCase inhibitors (Quizalofop, Fenoxaprop) and mitotic disruptors (Pendimethalin) to delay herbicide resistance in aggressive species such as *Trianthema* and *Echinochloa*.

1.9 Specialized Control Strategies for Problematic Perennial Weeds

- **Purple Nutsedge (*Cyperus rotundus* / Motha):**
  - *Growth Habit Persistence:* Underground tuber chains survive deep in the subsoil (15 to 40 cm), insulated from shallow cultivation. Tubers sprout repeatedly following each weeding round.
  - *Integrated Control Strategy:* Combine deep summer MB ploughing to bring tubers to the surface for solar drying with post-emergence directed applications of Glufosinate-ammonium 13.5% SL @ 2.5 to 3.0 L/ha or targeted spot applications of Glyphosate 41% SL (1%) using hooded nozzles along inter-row zones.

- **Bermudagrass (*Cynodon dactylon* / Doob ghas):**
  - *Growth Habit Persistence:* Deep rhizomes and prostrate stolons establish resilient sod mats.
  - *Integrated Control Strategy:* Summer soil inversion using chisel or disc ploughs. Apply Quizalofop-ethyl 5% EC @ 1000 mL/ha or Fenoxaprop-p-ethyl 9.3% EC @ 750 mL/ha when stolons produce fresh 3 to 4-leaf growth, followed by inter-row blade harrowing 10 days later.

- **Field Bindweed and Morning Glory (*Convolvulus arvensis* and *Ipomoea* spp.):**
  - *Growth Habit Persistence:* Twining vines that climb up cotton stalks, pulling branches downward and interfering with boll opening and picking.
  - *Integrated Control Strategy:* Apply pre-emergence Pendimethalin 30% EC @ 3.3 L/ha to suppress germinating vine seedlings. Manually untangle and uproot established vines during early flowering before stems lignify.

1.10 Integrated Weed Management (IWM) Chronological Master Roadmap

- **Chronological Seasonal Management Stages:**
  1. *Off-Season Summer Window (May to June):*

- Execute deep summer Mouldboard ploughing to invert soil clods.
- Leave soil exposed to intense solar heat for 2 to 3 weeks for thermal desiccation of perennial tubers and rhizomes (*Cyperus rotundus*, *Cynodon dactylon*).
- Uproot and burn alternate weed hosts (*Abutilon*, *Parthenium*, *Xanthium*) along bunds, irrigation channels, and field borders to eliminate whitefly and CLCuD reservoirs.

2. *Sowing and Pre-Emergence Window (0 to 3 DAS):*
   - Sow certified acid-delinted seeds in a firm, pulverized seedbed with optimal moisture (*Wattar*).
   - Apply pre-emergence herbicide Pendimethalin 30% EC @ 2.5 to 3.3 L/ha (or Pendimethalin 38.7% CS @ 1.75 L/ha) in 500 L water within 24 to 48 hours of sowing.
   - In Central and South India, intercrop short-duration cowpea or soybean in a 1:2 row ratio between cotton rows.
3. *Early Vegetative Window (15 to 30 DAS):*
   - Perform first intra-row manual hoeing and weeding with Kasola at 18 to 20 DAS prior to first post-sowing irrigation.
   - Operate animal- or tractor-drawn cultivators (Triphali) or rotary weeders across inter-row spaces to uproot weed flushes and break surface soil crusts.
   - If heavy broadleaf weeds (*Trianthema*, *Amaranthus*) or grassy weeds (*Echinochloa*) emerge after irrigation or rainfall, spray Pyrithiobac-sodium 6% + Quizalofop-ethyl 4% 10 MEC @ 1250 mL/ha in 375 L water, or apply Pyrithiobac-sodium 10% EC @ 625–750 mL/ha plus Quizalofop-ethyl 5% EC @ 1000 mL/ha.
4. *Grand Vegetative and Squaring Window (30 to 60 DAS):*
   - Execute second round of mechanical intercultivation with tractor-drawn Triphali at 35 to 40 DAS.
   - Perform second manual hand weeding along crop rows to remove intra-row weeds before canopy closure.
   - If stubborn inter-row weeds persist, apply directed hooded sprays of Paraquat dichloride 24% SL @ 1.5 L/ha or Glufosinate ammonium 13.5% SL @ 2.5 to 3.0 L/ha strictly between rows, avoiding cotton foliage.
   - Terminate mechanical tractor tillage once sympodial branches expand across inter-row spaces (beyond 60 DAS) to prevent root pruning and flower drop.
5. *Boll Development and Pre-Harvest Window (60 to 150+ DAS):*
   - Maintain field borders and water channels clean and weed-free to prevent late weed seed production.
   - Hand-pull large escaped weeds (such as *Abutilon*, *Xanthium*, *Ipomoea*) prior to boll opening to keep lint clean and prevent trash contamination during harvest.
   - Shred stalks in-situ and deep plough post-harvest to prevent weed seed burial and carryover.

- **Core Integrated Weed Management Operational Summary:**
  - *Critical Competition Window:* First 8 to 9 weeks after sowing (0 to 60 DAS).
  - *Preventive Cultural Foundations:* Summer MB ploughing with 2 to 3 weeks solarization; complete border weed sanitation (*Parthenium*, *Abutilon*); leguminous cover crop intercropping.
  - *Mechanical Strategy:* First intercultivation with tractor/animal Triphali and manual Kasola hoeing at 18 to 20 DAS; repeat 2 to 3 times before 60 DAS; discontinue intercultivation after sympodial flowering to avoid root pruning.
  - *Chemical Pre-Emergence Strategy:* Pendimethalin 30% EC @ 2.5 to 3.3 L/ha (or 38.7% CS @ 1.75 L/ha) in 500 L water within 48 hours of sowing onto moist, clod-free seedbeds.
  - *Chemical Post-Emergence Strategy:* Selective foliar spray of Pyrithiobac-sodium 10% EC (@ 625–750 mL/ha) for broadleaf weeds and Quizalofop-ethyl 5% EC (@ 1000 mL/ha) for grasses at 20 to 30 DAS; hooded inter-row sprays of Paraquat 24% SL (@ 1.25–2.0 L/ha) or Glufosinate 13.5% SL (@ 2.5–3.0 L/ha) at 45 to 60 DAS.